

POLYP DETECTION MEDICAL REPORT

Analysis Date: June 04, 2026

VIDEO INFORMATION

Video ID:	f04c62cf-982d-46f0-b840-f46d9116d344
Total Frames:	1046
Frame Rate:	25.0 fps (assumed)
Duration:	41.8 seconds

DETECTION SUMMARY

Detection Method	Count	Status
YOLO Detections	211	✓
RT-DETR Detections	590	✓
MedSAM2 Detections	658	✓
Consensus (3-Model Agreement)	3	✓
Risk Classification	Count	Percentage
HIGH RISK	0	0.0%
MEDIUM RISK	0	0.0%
LOW RISK	3	100.0%
TOTAL ANALYZED	3	100.0%

DETAILED POLYP ANALYSIS

POLYP #1 — LIFTED_POLYP [LOW_RISK] (Tracked across 24 frames)

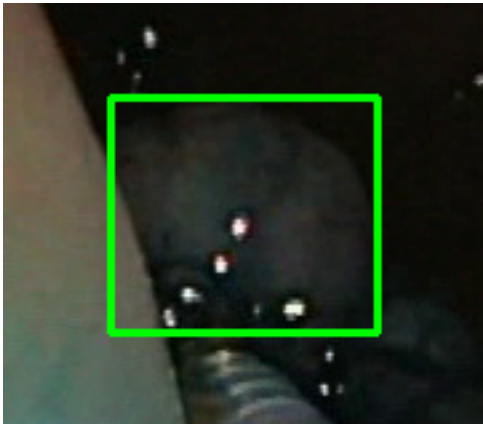


Feature	Value	Interpretation
Frame Index	12	Frame 12 of 1046
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 60.7%
Redness Score	0.017	Color-based vascularization
Relative Radius	19.2%	Polyp size as % of frame diagonal
Texture Score	0.124	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	91.9%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #2 — LIFTED_POLYP [LOW_RISK] (Tracked across 23 frames)



Feature	Value	Interpretation
Frame Index	85	Frame 85 of 1046
Detection Method	Detector Consensus (YOLO / RT-DETR)	Model Agreement (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 59.7%
Redness Score	0.000	Color-based vascularization
Relative Radius	14.2%	Polyp size as % of frame diagonal
Texture Score	0.010	Surface morphology
Vessel Visibility	0.000	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	NORMAL_MUCOSA	Normal mucosal pattern — no polyp morphology identified
Detection Confidence	83.5%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

POLYP #3 — LIFTED_POLYP [LOW_RISK] (Tracked across 21 frames)



Feature	Value	Interpretation
Frame Index	590	Frame 590 of 1046
Detection Method	Detector Consensus (MedSAM2 Unavailable)	Detector Consensus (MedSAM2 Unavailable) (MedSAM2 unavailable for this video)
POLYP TYPE	LIFTED_POLYP	Type confidence: 81.7%
Redness Score	0.035	Color-based vascularization
Relative Radius	12.1%	Polyp size as % of frame diagonal
Texture Score	0.576	Surface morphology
Vessel Visibility	0.013	Vascularization indicator
Risk Tier	LOW RISK	Validated by ESGE 2017 / NICE classification
Clinical Classification	HIGH_RISK	Final symbolic reasoning bucket
Expert Prediction	HIGH_RISK	Decision Tree output
Clinical Class	ADENOMATOUS_POLYP	Polypoid lesion — discrete mucosal protrusion identified
Detection Confidence	78.0%	YOLO / RT-DETR / track confidence

Type Assessment: Lifted lesion post-injection — submucosal plane confirmed. Resection feasible.

Low-confidence LOW RISK detection - Follow standard protocol

CLINICAL IMPRESSION

Summary:

A total of 3 polyps were detected and analyzed using multi-model consensus and AI-based risk stratification.

Risk Distribution:

- HIGH RISK polyps: 0 (0.0%)
- MEDIUM RISK polyps: 0 (0.0%)
- LOW RISK polyps: 3 (100.0%)

Recommendation:

All detected polyps are classified as LOW RISK. Standard surveillance protocol is recommended.

Technical Details:

- Detection: Multi-model consensus (YOLO, RT-DETR, MedSAM2)
- Features: 444-dimensional vectors (384 SSL + 60 biomarkers)
- Classification: Cluster-specific Decision Tree experts
- Confidence: Based on deep learning + medical heuristics

Disclaimer: This report is generated by an AI-assisted research system for academic and research purposes only. All clinical recommendations are derived from published ESGE 2017, NICE CG118/CG131, and BSG 2019 colonoscopy guidelines. This output does not constitute medical advice and must not be used as the sole basis for clinical decisions. All findings must be reviewed and confirmed by a qualified gastroenterologist or endoscopist.